👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 5 · STANDARD 6.GR.2

Volume of Rectangular Prisms

Lesson 5-10 · Teacher Copy

Name: _____ **Date:** _____ **Class:** _____

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can find the volume of a rectangular prism, including ones with fractional edge lengths, using base area × height.

Language Objective: I can explain my work using the words volume, rectangular prism, dimensions, and base area.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Volume · **Rectangular prism** · **Cubic units** · **Dimensions** · **Base area**

Level 3: try the blanks from memory first, then check the bank.

1. The amount of space inside a three-dimensional solid is its _____.

2. A solid with six rectangular faces is a _____.

3. The unit used to measure volume is _____.

4 The measurements of length, width, and height of a solid are its _____ .

5 The area of the bottom face of a prism, multiplied by height to find volume, is the _____ .

Write About the Math

How many gallons of water does the small tank hold?
¿Cuántos galones de agua tiene la pecera pequeña?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Volume** *Volumen*
- ☐ **Rectangular prism** *Prisma rectangular*
- ☐ **Cubic units** *Unidades cúbicas*
- ☐ **Dimensions** *Dimensiones*
- ☐ **Base area** *Área de la base*

Model: *You still multiply length × width × height even when an edge is a fraction or decimal.* — Students explain the formula stays the same ($V = l \times w \times h = 2 \times 1.5 \times 1 = 3$ ft^3) and that a fractional edge just means multiplying with a fraction or decimal.

Start

Complete the frames. Use the word bank.
Completa las oraciones. Usa el banco de palabras.

- **The tank's volume is ___ in³ because V = ___ × ___ × ___. It holds about ___ gallons because ___ ÷ 231 ≈ ___.**
- **My answer is ___ because ___.**
  Mi respuesta es ___ porque ___.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ___.**
  Mi afirmación es ___.
- **My evidence is ___, so ___.**
  Mi evidencia es ___, entonces ___.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.

- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Volume
2. **Notes 2:** Rectangular prism
3. **Notes 3:** Cubic units
4. **Notes 4:** Dimensions
5. **Notes 5:** Base area
6. **Watch for this mistake:** *A common mistake in Volume of Rectangular Prisms is rounding a fractional edge length to the nearest whole number before multiplying, instead of using the exact fraction or decimal. For a box that is 2 ft × 1.5 ft × 1 ft, a student might round 1.5 down to 1 and compute 2 × 1 × 1 = 2 ft³, but the correct volume uses the exact edge length: 2 × 1.5 × 1 = 3 ft³ — a full cubic foot larger. Before you submit, ask: 'Did I multiply using the exact fractional or decimal edge length, or did I round it off first?'*
7. **Try It 1:** B. Box A, because 50 cubic in is more than 48 cubic in — *Box A: 5 × 5 × 2 = 50 cubic in. Box B: 4 × 4 × 3 = 48 cubic in. 50 > 48, so Box A holds more.*
8. **Try It 2:** C. 18 cubic in — *V = 4 × 3 × 1.5 = 12 × 1.5 = 18 cubic in. A fractional edge does not change the steps.*